Adaptive Edge-Aware Semantic Interaction Network for Salient Object Detection in Optical Remote Sensing Images

Xiangyu Zeng, Mingzhu Xu, *Member, IEEE*, Yijun Hu, Haoyu Tang, *Member, IEEE*, Yupeng Hu, *Member, IEEE*, and Liqiang Nie, *Senior Member, IEEE*

***Abstract*— In recent years, the task of salient object detection in optical remote sensing images (RSI-SOD) has received extensive attention. Benefiting from the development of deep learning, much progress has been made in RSI-SOD field. However, existing methods still face challenges in addressing various issues present in optical RSI, including uncertain numbers of salient objects, cluttered backgrounds, and interference from shadows. To address these challenges, we propose a novel approach, adaptive edge-aware semantic interaction network (AESINet) for efficient salient object detection (SOD). Specifically, to improve the extraction of complex edge information, we design a local detail aggregation module (LDAM). This module can adaptively enhance the edge information of salient objects by leveraging our proposed difference perception mechanism. Notably, our difference perception mechanism is a novel edge enhancement method without the supervision of edge ground truth. Additionally, to accurately locate salient objects of varying numbers and scales, we design a multiscale feature extraction module (MFEM), which effectively captures and utilizes multiscale information. Moreover, we design the deep semantic interaction module (DSIM) to identify salient objects amidst cluttered backgrounds and effectively mitigate the interference of shadows. We conduct extensive experiments on three well-established optical RSI datasets and the results demonstrate that our proposed model outperforms 14 state-of-the-art methods. All codes and detection results are available at https://github.com/xumingzhu989/AESINet-TGRS.**

***Index Terms*— Adaptive edge enhancement, deep semantic interaction, optical remote sensing images, salient object detection (SOD).**

Manuscript received 7 December 2022; revised 24 May 2023; accepted 1 July 2023. Date of publication 1 August 2023; date of current version 16 August 2023. This work was supported in part by the Key R&D Program of Shandong Province, China (Major Scientific and Technological Innovation Projects) under Grant 2022CXGC020107; in part by the National Natural Science Foundation of China (NSFC) under Grant 62206157, Grant 62276155, and Grant 62206156; and in part by the Natural Science Foundation of Shandong Province under Grant ZR2022QF047. *(Corresponding author: Mingzhu Xu.)*

Xiangyu Zeng is with the School of Software, Shandong University, Jinan 250101, China, and also with the School of Computer Science and Technology, Shandong University, Qingdao 266237, China (e-mail: z15264367990@163.com).

Mingzhu Xu, Yijun Hu, Haoyu Tang, and Yupeng Hu are with the School of Software, Shandong University, Jinan 250101, China (e-mail: xumingzhu@sdu.edu.cn; juneyee2@126.com; tanghao258@sdu.edu.cn; huyupeng@sdu.edu.cn).

Liqiang Nie is with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: nieliqiang@gmail.com).

Digital Object Identifier 10.1109/TGRS.2023.3300317

I. Introduction

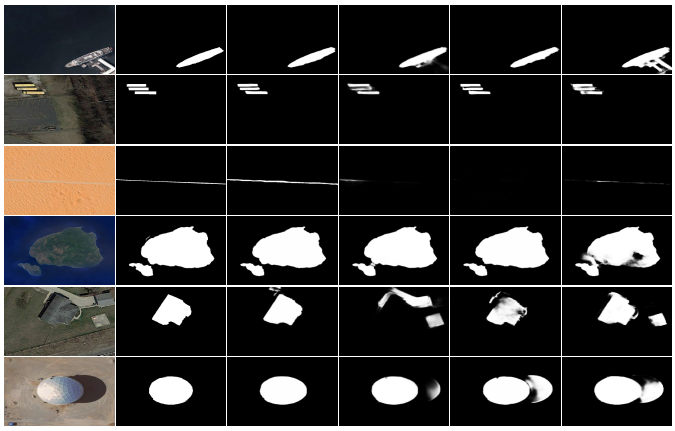
SALIENT object detection (SOD) is a task to identify the most visually appealing objects/regions in images/videos and generate pixel-level saliency maps [1], [2], [3]. Recent advancements in deep neural networks have led to remarkable achievements in SOD for natural scene images (NSIs) [4], [5], [6], [7], [8]. Concurrently, researchers have begun to focus on SOD in optical remote sensing images (RSI-SOD) [9], [10], [11], [12], [13], [14], [15], [16], [17]. RSI-SOD involves locating the most attractive and category-independent objects, serving as a pre-processing step to enhance the information processing in various classical remote sensing image tasks, such as target extraction [18], super-resolution [19], image fusion [20], and scene classification [21]. Different from SOD in NSIs, there are many challenges in RSI-SOD that have not been well addressed. It should be noticed that applying the NSI-SOD methods directly to the RSI-SOD often fails to produce satisfactory results (as shown in column (f) of Fig. 1). This may be attributed to the specific imaging modality of optical remote sensing images, which are captured at high altitudes by satellite or aerial sensors [22]. Typically, optical remote sensing images cover a wide range of diverse objects and complex backgrounds. Therefore, the research on RSI-SOD is a more challenging task, which is necessary to be further explored.

In recent years, the advancement of convolutional neural networks has significantly propelled the progress in RSI-SOD, and researchers have proposed many excellent algorithms [9], [10], [11], [12], [13], [14], [15], [16], [17]. A notable one is multi-scale joint region and boundary model (MJRBM) [12], which introduces a joint learning framework that simultaneously optimizes boundary features and salient regions to generate high-quality saliency maps. Besides, edge-guided recurrent positioning network (ERPNet) [16] employs two different decoders. One is designed to detect salient objects, while the other focuses on detecting the edges of salient objects that are adopted to improve the prediction accuracy of salient objects.

However, despite the progress made by the existing methods, several challenges persist in effectively processing RSIs. First, the edges of salient objects in RSIs are more intricate compared to those in regular NSIs. Optical RSIs frequently exhibit complex edges in objects such as islands and houses

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f)

Fig. 1. Performance comparison of the advanced RSI-SOD and NSI-SOD methods and our method under some optical RSIs. (a) Optical RSIs. (b) GT. (c) Our method. (d) MJRBM (RSI-SOD Method) [12]. (e) ERPNet (RSI-SOD Method) [16]. (f) gated network (GateNet) (NSI-SOD Method) [23].

(e.g., the fourth and fifth rows in Fig. 1). While methods like edge-aware multiscale feature integration network (EMFINet) [11], MJRBM [12], and HFANet [17] have effectively addressed the impact of complex edges of salient objects on detection. The above methods adopt the edge enhancement method of edge truth supervision. We find that unsupervised adaptive edge enhancement method remains to be explored. Therefore, our aim is to propose an adaptive edge enhancement method that does not rely on edge ground truth (GT) for supervision. Second, the number of salient objects varies greatly. Due to the extensive imaging coverage of optical RSI and the random shooting scene, there may be multiple salient objects (e.g., the second row of Fig. 1), each exhibiting large-scale differences. RSIs captured at different heights and environments may encompass objects ranging from a few pixels (e.g., a car) to covering a substantial portion of the image (e.g., an island) (e.g., the scale difference between the second and fourth rows in Fig. 1). In addition, the location of salient objects in RSIs tends to be more random. Since NSI is typically captured by handheld cameras, salient objects are more likely to appear in the central area of the image. However, optical RSI is captured by satellites or drones, with salient objects appearing randomly, including the edge of the image (e.g., the first and second rows in Fig. 1). Furthermore, optical RSIs have cluttered backgrounds. The imaging environment of the optical RSIs results in all objects in the scene appearing equally clearly in the image, leading to rich and cluttered background details (e.g., the fifth row of Fig. 1). Lastly, salient objects are sometimes accompanied by shadows. This situation is common in some images with houses, where shadows are often mistakenly identified as part of the salient objects (e.g., the sixth row of Fig. 1).

To address the aforementioned challenges, we propose a novel model named adaptive edge-aware semantic interaction network (AESINet), which consists of three key modules: the local detail aggregation module (LDAM), the multiscale feature extraction module (MFEM), and the deep semantic interaction module (DSIM). To effectively capture the intricate edges of salient objects, we design a LDAM. Our aim is to propose an unsupervised edge enhancement method that eliminates the need for edge GT supervision. The core of LDAM lies in our proposed difference perception mechanism, which adaptively extracts the edges of salient objects through the difference between average pooling and max pooling feature maps. To effectively adapt to different sizes and numbers of salient objects, we design a MFEM. This module extends the deep information of the encoder to different scales, and performs feature extraction through multiple parallel dilated convolutions. It allows the model to adapt more effectively to diverse object sizes. To mitigate the interference of shadows and cluttered backgrounds on SOD, we design a DSIM. This module projects the deep information of the encoder into three graphs, and performs deep semantic feature inference through three parallel graph convolutions. This enables the model to accurately identify the location of salient objects from cluttered backgrounds.

In summary, the contributions of this article are as follows.

1) We propose a novel AESINet. We apply DSIM for semantic feature inference, enabling accurate identification of salient objects, and apply MFEM to achieve further precise localization of salient objects. Moreover, the edge details of salient objects are enhanced by LDAM. Through extensive evaluation on three challenging datasets, our method outperforms 14 state-of-the-art methods, highlighting the superiority of our proposed method.
2) We propose a LDAM. By sensing the difference between the feature maps of the two branches, effective edge information can be extracted, and then the edge features of salient objects can be enhanced adaptively. Different from the other methods that rely on edge GT for supervision, our unsupervised adaptive edge extraction method eliminates the supervision of edge GT in RSI-SOD.
3) We propose a MFEM. This module comprehensively constructs multiscale dilated convolution, which can effectively extract multiscale salient object features and then achieve accurate localization of salient objects.
4) We propose a DSIM. This module projects pixels as graph vertices, and performs graph convolution operations in different interaction domains. It can effectively extract global context information at different scales to identify salient objects.

II. Related Work

In this section, we will introduce the classical related research on NSI-SOD and RSI-SOD, respectively.

A. SOD in Natural Scene Images

SOD first appeared in the field of NSIs. The earliest significant achievements in this field were the traditional NSI-SOD methods. These methods are unsupervised learning and mainly rely on traditional image processing methods to extract interpretable artificial features, including saliency tree [24], random walk [25], [26], [27], structured matrix decomposition [28], and two-layer sparse graph [29]. Different from the aforementioned unsupervised learning methods,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

some researchers introduced innovative methods based on supervised/semi-supervised learning [30]. Although the traditional methods mentioned earlier did not achieve satisfactory performance, they nonetheless presented numerous innovative perspectives that have contributed fresh ideas for subsequent investigations in CNN-based deep learning methods.

In recent years, convolution-based deep learning methods have greatly promoted the development of NSI-SOD, and many classic methods have been proposed by distinguished researchers. Many of these methods [6], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43] are inspired by classic computer vision, such as Visual Geometry Group (VGG) and ResNet, resulting in the adoption of encoder–decoder architecture. Moreover, the mainstream CNN-based methods are pixel-level models. To enhance the detection performance of salient objects, various methods have been explored by researchers, including global context-aware aggregation [31], [32], attention mechanism [6], [32], [33], [34], edge guidance [5], [38], multiscale interaction [4], and deep supervision [39]. Different from traditional attention mechanisms, Chen et al. [34] proposed a side-output residual learning method based on reverse attention. This approach enables the model to detect and fill in the missing parts of salient objects without raising concerns about redundancy in current detections. Zhang et al. [35] aggregated all five-layer outputs of the encoder into five different resolutions, respectively, and applied deep supervision to multiple high-resolution side outputs. In addition, Zhang et al. [36] introduced a gated bidirectional message-passing module to facilitate deep interaction between deep semantic information and shallow spatial details. Qin et al. [37] developed a novel residual refinement module and proposed the structure similarity (SSIM) loss. Although the traditional NSI-SOD methods may not fully address challenges in the field of RSI-SOD, they have greatly inspired researchers and generated numerous novel ideas for RSI-SOD research.

B. SOD in Optical Remote Sensing Images

As a result of the rapid development of CNN-based NSI-SOD architectures, many outstanding researchers have combined these architectures with traditional RSI-SOD methods, leading to the development of several noteworthy models. Li et al. [9] published a challenging dataset (i.e., ORSSD) and proposed an L-V-shaped network (i.e., LVNet) dedicated to RSI-SOD. LVNet is one of the first models to introduce CNN into the field of RSI-SOD and demonstrated versatility across various optical RSI scenarios. Shortly after, Zhang et al. [10] extended ORSSD and released the more comprehensive and diverse dataset EORSSD. They introduced the attention mechanism to global context modeling, proposing DAFNet. Zhou et al. [11] proposed EMFINet, an encoder–decoder architecture network that integrates deep feature aggregation, multiscale feature extraction, and edge information mining. Tu et al. [12] released a more comprehensive dataset ORSI and proposed a brand-new model MJRBM. They innovatively construct two sets of encoders and decoders, leveraging the interaction between them to enhance SOD. Cong et al. [14] introduced the relational reasoning module into the RSI-SOD framework for the first time, and combined the multiscale convolution and attention mechanism to design a parallel module to improve the detailed information. Zhou et al. [16] established two distinct decoders, one for salient edge detection and the other for SOD, where the detected saliency edges enhance SOD. Li et al. [13] applied multiple attention mechanisms comprehensively, and combined the complementarity of foreground information, edge information, background information, and image-level global information effectively, thereby making multiscale information extraction more comprehensive. In addition, Li et al. [15] proposed adjacent context coordination network (ACCoNet), which establishes the feature interaction between adjacent blocks in the encoder, and implemented a bifurcated aggregation structure to the decoder. Recently, Wang et al. [17] proposed a novel RSI-SOD architecture named HFANet. They proposed a hybrid encoder combining CNN and transformer, applied a feature alignment mechanism between layers, and introduced an interactive loss function for edges and salient objects.

However, several challenges in RSI-SOD are still not well addressed by the aforementioned methods, such as complex edges, random number of objects, cluttered background, and shadow interference. Therefore, to overcome these challenges, we propose a novel AESINet, which will be detailed in Section III.

III. Methodology

In this section, we detail the proposed AESINet. As shown in Fig. 2, the model adopts a classical encoder-decoder architecture. The traditional VGG16 architecture is used as the encoder of the model. In addition, we design the DSIM that facilitates the extraction of relevant global contextual information at different levels, enabling more profound semantic reasoning. This approach effectively mitigates the risk of detecting irrelevant objects and improves the accuracy of SOD. Furthermore, to effectively capture salient objects with uncertain sizes and numbers, we apply the MFEM to the deep output of the encoder. MFEM achieves this by scaling the feature maps to three distinct sizes and expanding the receptive field using diverse dilated convolutions in a parallel structure. Finally, we apply the LDAM to the shallow output of the encoder. By sensing the difference information appearing between the two-branch processing paths, this module can adaptively complement the edge details of salient objects. To summarize, three different modules are employed in a sequential manner, moving from deep to shallow, effectively representing three stages of SOD, and progressing from coarse to fine. First, DSIM roughly localizes salient objects. Then, MFEM detects the number and scale of salient objects. Finally, LDAM complements the edge details of salient objects.

A. Encoder and Decoder

Following other researchers [10], [11], [12], [15], [16], we use VGG16 [44] as the encoder for the model. The input image size of the encoder is $224 \times 224 \times 3$. As shown in Fig. 2, our encoder has five outputs, defined as $f_i \in \mathbb{R}^{C_i \times H_i \times W_i} (i \in \{1, 2, 3, 4, 5\})$. $C_{1,2,3,4,5} = 64, 128, 256, 512, 512$ and $H_{1,2,3,4,5} = 224, 112, 56, 28, 14$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

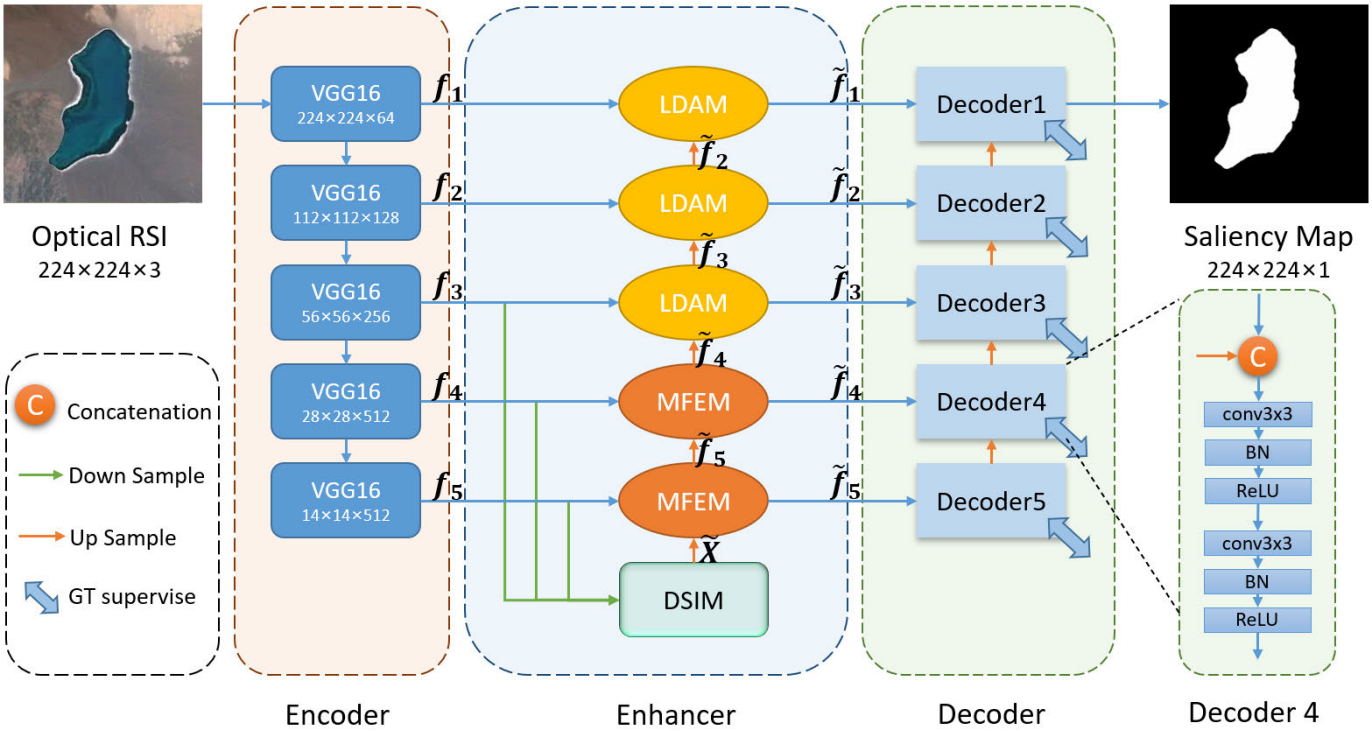


Fig. 2. Overall architecture of our network. The network can be divided into five parts: encoder, LDAM, MFEM, DSIM, and decoder. First, the encoder performs feature extraction on the optical RSI to obtain five outputs. The deep features extracted by the encoder are fed into DSIM, where reasoning is performed on the deep semantic features to identify salient objects. Next, MFEM will perform multiscale information extraction to accommodate different numbers and sizes of salient objects. Then, the LDAM located in the shallow layer enhances the detailed features such as edges through our designed adaptive method. Finally, the decoder aggregates feature maps from different layers and outputs the saliency map. During the training phase, we apply GT supervision to all five layers of the decoder.

For the decoder, we use two layers of basic convolutions, with each layer consisting of convolution 3 × 3, batch normalization, and rectified linear unit (ReLU). To aggregate the information among different layers, from the first to the fourth part of our decoder, we dimensionally link the output of the deep part with the input of this part. The specific details of the decoder are shown on the right side of Fig. 2.

B. Local Detail Aggregation Module (LDAM)

The extraction of detailed information is crucial to the entire RSI-SOD architecture. Regarding the complex edges of salient objects, how to effectively extract details is a major challenge in this field, since rich detailed information is in the shallow outputs of the encoder. To make full use of the detailed information existing in the shallow layer, we design the LDAM, as shown in Fig. 3.

Assume that the two inputs of the ith LDAM are f_i and $\widetilde{f}_{i+1}$, respectively, where f_i is the side-output feature map of the ith layer in the encoder, and $\widetilde{f}_{i+1}$ is the feature map of the $i+1$th layer processed by the enhancer. Obviously, considering that the size of feature map in the ith layer is $H \times W$, then $f_i \in \mathbb{R}^{C_i \times H \times W}$ and $\widetilde{f}_{i+1} \in \mathbb{R}^{C_{i+1} \times (1/2)H \times (1/2)W}$.

For simplicity, we define the spatial attention operation and the channel attention operation as follows:

$$F_{\mathrm{CA}}(x) = x + x \odot \mathrm{CA}(x) \tag{1}$$

$$F_{\mathrm{SA}}(x_1, x_2) = x_1 + x_1 \otimes \mathrm{SA}(x_2) \tag{2}$$

where $\odot$ is the multiplication of the channel dimension, $\otimes$ refers to the multiplication of the space dimension, and $+$ denotes element-wise addition. CA() is the channel attention module, consisting of global max pooling (channel dimension), fully connected layer × 2 and Sigmoid. SA() is the spatial attention module, consisting of global max pooling (spatial dimension), Conv1 × 1, and Sigmoid.

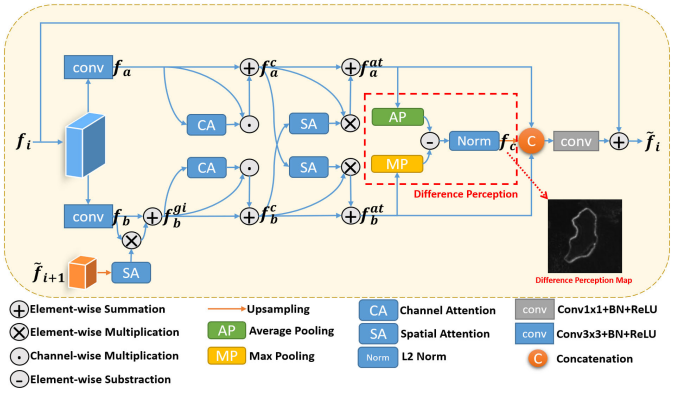


Fig. 3. Internal structure of LDAM. This module acquires effective edge information by sensing the difference information between the two branches, and adaptively enhances edge features.

We apply two 3 × 3 convolutions to f_i to construct a two-branch process

$$f_a = \mathrm{Conv}_1(f_i), \quad f_b = \mathrm{Conv}_2(f_i) \tag{3}$$

where $\mathrm{Conv}_1()$ and $\mathrm{Conv}_2()$ represent two basic convolution modules, comprising 3 × 3 convolution, Batch Norm, and ReLU.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

Next, we apply deep informative guidance to f_b in the form of spatial attention. Specifically, we upsample $\widetilde{f}_{i+1}$ to $H \times W$ and input it into the spatial attention module, processing f_b in the form of residual links, i.e.,

$$f_b^{\mathrm{gi}} = F_{\mathrm{SA}}\left(f_b, \widetilde{f}_{i+1}\right). \tag{4}$$

As shown in Fig. 3, we apply channel attention and spatial attention to the two-branch feature maps after convolution, i.e.,

$$f_a^c = F_{\mathrm{CA}}(f_a), \quad f_b^c = F_{\mathrm{CA}}\left(f_b^{\mathrm{gi}}\right) \tag{5}$$

$$f_a^{\mathrm{at}} = F_{\mathrm{SA}}\left(f_a^c, f_b^c\right), \quad f_b^{\mathrm{at}} = F_{\mathrm{SA}}\left(f_b^c, f_a^c\right). \tag{6}$$

After the aforementioned operations, we perform difference perception processing on the feature maps of the two branches. Specifically, we apply max pooling and average pooling (both in spatial dimension) to the feature maps of the two branches, respectively. And then we find the distance between the features of the corresponding spatial positions of the two feature maps. Thereafter, a $1 \times H \times W$ difference perception map is obtained. This process can be expressed as

$$f_c = L2_{\mathrm{norm}}\left(\mathrm{AP}\left(f_a^{\mathrm{at}}\right) - \mathrm{MP}\left(f_b^{\mathrm{at}}\right)\right) \tag{7}$$

where AP() and MP() represent average pooling and max pooling, $-$ is element-wise subtraction, $L2_{\mathrm{norm}}()$ solves out the L2norm of the channel dimension, and f_c is the difference perception map obtained.

The aforementioned operations are verified as effective. The difference perception mechanism utilizes the properties of average pooling and max pooling. Average pooling focuses on extracting smooth foreground and background information, while max pooling extracts sharpened edge information. Subtracting the results of average pooling from max pooling can better remove foreground and background noise regarding the sharpened edges, thus obtaining purified edge features.

In addition, the difference perception mechanism is explored from the perspective of more basic computational processes. The edges of salient objects appear as spatially adjacent features with relatively drastic changes on the feature map. After two pooling operations, we extract the maximum and average values within a tiny region of each 2×2 pixel on two feature maps. If there are relatively drastic changes in a certain area, the difference between the maximum value and the average value in the tiny area of 2×2 pixels is relatively large. On the contrary, for the parts that are not edges, the features are relatively smooth, and the difference among adjacent pixels is rather small. In addition, through learning, the feature maps of the two branches will magnify this difference, further improving the performance. Fig. 4 shows the principle and effect of this operation, where the closer each pixel is to white, the higher the value. Case 1 is a schematic of the edge area, and the features of adjacent pixels change drastically. Case 2 is a schematic of a non-edge area, and the features of adjacent pixels change very little.

Different from traditional supervised edge enhancement methods [10], [11], [12], [13], [16], [17], this method is a novel edge-aware method without supervised by the edge GT map explicitly. It can sense the edge information of the mutation in the feature map and naturally integrate it into the local details

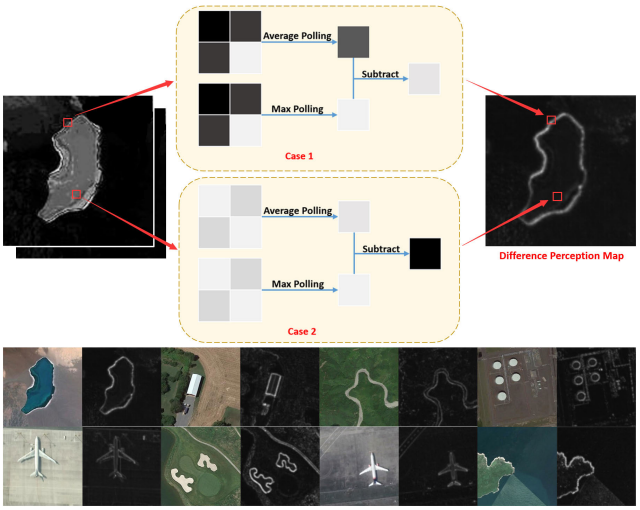


Fig. 4. Upper part shows how the difference perception mechanism works. The closer the pixel block is to white, the larger the value. The bottom half shows a portion of the input image and the corresponding difference perception map. The method does not need to supervise the edge, and can adaptively extract edge information. From the visualization results point of view, the effectiveness of our proposed method is verified.

to enhance its edge features. Therefore, this method provides a novel perspective on edge enhancement.

Finally, we concat the obtained dual-branch feature map with the difference perception map, perform adaptive aggregation through 1×1 convolution, and then obtain the output $\widetilde{f}_i$ of LDAM, i.e.,

$$\widetilde{f}_i = \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(f_a^{\mathrm{at}}, f_b^{\mathrm{at}}, f_c\right)\right) + f_i \tag{8}$$

where $+$ is element-wise addition. Because we embed LDAM into the architecture in the form of residual links, we need to add f_i to the feature map processed by LDAM.

The validation of the effectiveness of LDAM is presented in the subsection of the ablation studies in the experiment section (located in Section IV-C2).

C. Multiscale Feature Extraction Module (MFEM)

Different salient objects in optical RSI differ in size, and naive CNN is not well suited for objects of different sizes. Some existing models [11], [12], [15], [17] extract multiscale information through multiple dilated convolutional layers computed in parallel. Dilated convolutions with different dilation rates can indeed play a better role in the extraction of multiscale information, but researchers often ignore the influence of different sizes of feature maps on multiscale information extraction. Toward this end, we propose a new module, named MFEM, which can extract effective multiscale information in different scales of the feature map, as shown in Fig. 5.

First, we scale the input feature maps $f_i \in \mathbb{R}^{C_i \times H \times W}$ to three different scales: $f_u \in \mathbb{R}^{C_i \times 2H \times 2W}$, $f_o \in \mathbb{R}^{C_i \times H \times W}$, and $f_d \in \mathbb{R}^{C_i \times (1/2)H \times (1/2)W}$, i.e.,

$$f_u = f_i \uparrow, \quad f_o = f_i, \quad f_d = f_i \downarrow \tag{9}$$

where $\uparrow$ means $2\times$ spatial upsampling operation and $\downarrow$ means $2\times$ spatial downsampling operation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

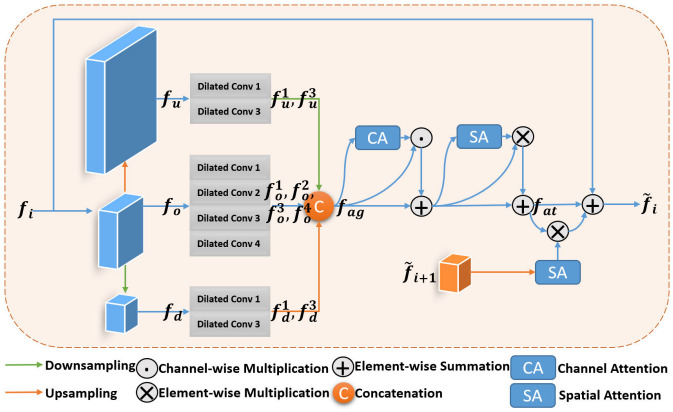


Fig. 5. Internal structure of MFEM. This module effectively extracts multiscale features of salient objects by scaling the feature maps to different scales, making it better adaptive to the interference caused by scale changes to the model.

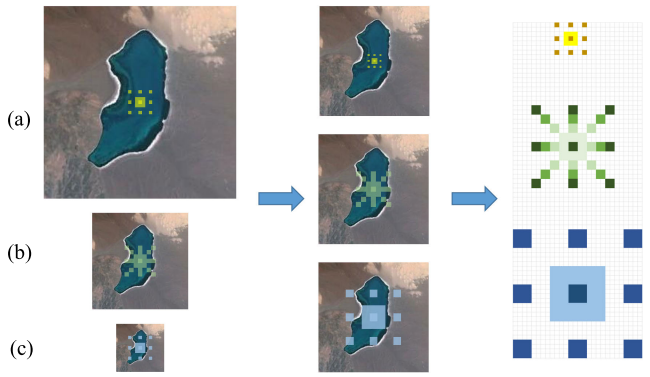


Fig. 6. Receptive fields of the three branches of MFEM. A: Upsampling branch, the corresponding receptive field is the yellow part. B: The middle branch, the corresponding receptive field is the green part. C: Downsampling branch, the corresponding receptive field is the blue part.

Then, the three feature maps f_u, f_o, and f_d are respectively subjected to different dilation convolutions in parallel. f_o undergoes four dilation convolutions with dilation rates of 1, 2, 3, and 4, respectively. f_u and f_d go through dilated convolutions with dilation rates of 1 and 3, i.e.,

$$f_u^k = \mathrm{DiConv}_k(f_u), \quad k \in \{1, 3\} \tag{10}$$

$$f_o^k = \mathrm{DiConv}_k(f_o), \quad k \in \{1, 2, 3, 4\} \tag{11}$$

$$f_d^k = \mathrm{DiConv}_k(f_d), \quad k \in \{1, 3\} \tag{12}$$

where $\mathrm{DiConv}_k()$ is a dilated convolution with dilation rate k. It should be noted that each dilated convolution reduces the channel dimension of the feature map from C_i to $(1/8)C_i$.

The receptive fields of our method are different from traditional parallel convolutions. Conventional parallel convolution can only achieve the same effect as B. The upsampling and downsampling branches (A and C) can bring two additional distinct receptive fields. It is worth mentioning that our method not only makes the dilation rate of the convolution kernel different, but also makes the image area covered by a single pixel on the convolution kernel different. For example, suppose each kernel element of A has coverage of 1×1 pixel, each kernel element of B has coverage of 2×2 pixels, and each kernel element of C has coverage of 4×4 pixels (as shown on the right side of Fig. 6). Overall, the receptive field of our designed three-branch structure is more diverse. This is reflected in the overall scale of the convolution kernel and the coverage of each convolution kernel element. Therefore, our designed MFEM enables the model with remarkable multiscale feature extraction capabilities.

We concat the obtained eight feature maps by the channel dimension, i.e.,

$$f_{\mathrm{ag}} = \mathrm{Concat}\left(f_u^1 \downarrow, f_u^3 \downarrow, f_o^1, f_o^2, f_o^3, f_o^4, f_d^1 \uparrow, f_d^3 \uparrow\right). \tag{13}$$

Next, we apply the attention mechanism to f_{ag}, i.e.,

$$f_{\mathrm{at}} = F_{\mathrm{SA}}\left(F_{\mathrm{CA}}(f_{\mathrm{ag}}), F_{\mathrm{CA}}(f_{\mathrm{ag}})\right) \tag{14}$$

where $F_{\mathrm{SA}}()$ and $F_{\mathrm{CA}}()$ are detailed in (1) and (2).

Finally, we take the high-level feature $\widetilde{f}_{i+1}$ to spatially guide the features of this layer, i.e.,

$$\widetilde{f}_i = F_{\mathrm{SA}}(f_{\mathrm{at}}, \widetilde{f}_{i+1}) + f_i \tag{15}$$

where $+$ is element-wise addition. Since we embed MFEM into the architecture in the form of residual links, we need to add f_i to the feature map processed by MFEM.

The validation of the effectiveness of MFEM is presented in the subsection of the ablation studies in the experiment section (located in Section IV-C3).

D. Deep Semantic Interaction Module (DSIM)

Previous studies have verified that deep semantic information is crucial for RSI-SOD [10], [17]. However, for deep semantic information extraction, parallel convolutional combinations such as atrous spatial pyramid pooling (ASPP) are commonly used [15], [17]. CNN has a good local receptive field in shallow layers, but with a weak ability to extract long-range context in deep layers. In contrast, graph convolutional network (GCN) has a good ability to extract long-distance context using graph convolution to perform interactions among different vertices. Thereby, we propose an improved GCN method to extract deep semantic information. In addition, researchers usually only extracted context information for the single-layer output of the encoder, ignoring the correlation among the outputs of different blocks of the encoder. Therefore, we fully explore the correlations and concatenate the outputs of the deepest three blocks of the encoder to model the multiscale context in the output information.

The processing flow of DSIM is shown in Fig. 7. Each pixel of the aggregated feature map will be projected into two distinct graph vertices, representing disparate semantics, that is, whether a pixel contains salient objects or not. To extract context information at different scales to achieve more effective semantic feature inference, three parallel graph convolution processes are constructed. Specifically, as shown in the left part of Fig. 7, we construct three different connected graphs, where the graph vertices are connected within one pixel grid, 2×2 pixel grid, and all 7×7 pixel grid, respectively. Thereafter, the three parallel graph convolutions are conducted by aggregating information from different receptive fields. Finally, the three graph embedding features are weighted and transformed back into feature maps through reprojection. These operations are elaborated in the following.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

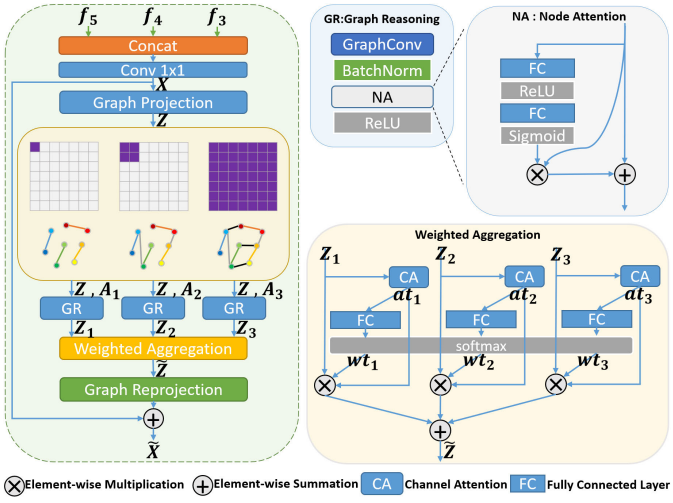


Fig. 7. Internal structure of DSIM. This module converts feature maps into graph structures through a specially designed graph projection method, and performs deep semantic inference through three parallel graph convolution branches, which can accurately identify salient objects from a semantic perspective.

1) Aggregation of Feature Maps: We downsample the outputs of the deepest three blocks from the encoder to 7×7 size, and then concat them in the channel dimension. Afterward, we perform dimensionality reduction through 1×1 convolutional layers. The channel dimension of the feature map is reduced from 1280-D to 512-D. We name the feature map after dimensionality reduction as $X \in \mathbb{R}^{512\times7\times7}$.

2) Graph Projection: First, each pixel of X is projected onto two graph vertices. Thereafter, we get the graph vertex information matrix $Z \in \mathbb{R}^{512\times98}$, which contains the information of 98 ($7 \times 7 \times 2$) vertices, and the information of each vertex is a 512-D vector.

The number of graph vertices corresponding to a single pixel can be specified by hyperparameters. Each graph vertex corresponds to a learnable anchor feature (trainable parameter). We determine the weights of different graph vertices by calculating the similarity between pixel features and anchor features. Then we calculate the information of each vertex based on vertex weights, pixel features, and anchor features. For a more detailed calculation process and theoretical explanation, please refer to [45]. Through training, different anchors corresponding to a pixel represent diverse intrinsic characteristics of the region. When the number of graph vertices is specified as 2, the information of two graph vertices projected by a particular pixel is expressed as the confidence of whether the pixel contains a salient object. One graph vertex indicates salient objects in the area, and the other graph vertex indicates no salient objects in the area. We conduct an experimental analysis of the performance difference brought by the number of graph vertices. The results show that setting the number of graph vertices to 2 can bring better performance with better computational efficiency. The specific experimental analysis is located in Section IV-C4.

3) Edge Construction: We construct a adjacency matrix $A \in \mathbb{R}^{98\times98}$ by calculating the similarity among vertex features (i.e., $A = Z^T Z$) The element a_{ij} in the adjacency matrix is the cosine similarity between z_i and z_j, that is, the edge weight from vertex i to vertex j in the graph.

We equally divide X into several blocks, and there are three ways to divide blocks for three different graphs. The sizes of the three blocks are one pixel, 2×2 pixels, and the entire feature map.

Next, we process the edges between the vertices of the graph according to how the three blocks are divided. We only keep the edges connected by graph vertices that belong to the same block. And, for the edges between graph vertices within the same block, we also delete edges with weights less than a certain threshold. In this way, we restrict the influence of graph convolution operations to three different scales and avoid vertices with lower similarity from interfering with each other. We get three different adjacency matrices, respectively as A_1, A_2, and A_3.

4) Graph Reasoning: We then perform graph reasoning on three graphs with different spatial relationships. The process is described as

$$Z_i = \mathrm{ReLU}(\mathrm{NA}(\mathrm{BN}(G_{\mathrm{conv}}(Z, A_i)))), \quad i \in \{1, 2, 3\} \tag{16}$$

where $G_{\mathrm{conv}}()$ is the graph convolution module provided by the PyTorch Geometric (PYG) library, BN() is batch normalization, NA() is the node attention, and the specific structure of NA is shown in Fig. 7.

5) Weighted Aggregation: We enhance the channel information of the three graphs by calculating the channel attention, and then calculating the weight of each graph according to the three channel attention maps as

$$\mathrm{at}_i = \mathrm{CA}(Z_i), \quad i \in \{1, 2, 3\} \tag{17}$$

$$wt_i = \frac{\exp(\mathrm{Liner}(\mathrm{at}_i))}{\sum_{i=1}^{3} \exp(\mathrm{Liner}(\mathrm{at}_i))}, \quad i \in \{1, 2, 3\} \tag{18}$$

where $\mathrm{at}_i \in \mathbb{R}^C$ is the channel attention map of the graph, and $wt_i \in \mathbb{R}^1$ is the weight of each graph. Linear() is a linear layer that reduces the input vector from C dimensions to 1-D.

Finally, we perform a weighted aggregation on the three graphs as

$$\widetilde{Z} = \sum_{i=1}^{3} wt_i \mathrm{at}_i \odot Z_i \tag{19}$$

where $\widetilde{Z}$ is the graph after aggregation and $\odot$ is the multiplication of the channel dimension.

6) Graph Reprojection: We follow the inverse process of graph projection to reproject the graph back to the feature map. Through reprojection, we restore the graph back to the feature map, and then residual link with X to get $\widetilde{X}$. X is the original feature map obtained by operation *1)*, and $\widetilde{X}$ is the output of DSIM. Feature map $\widetilde{X}$ can further be applied to the fifth layer of MFEM as high-level guidance.

The validation of the effectiveness of DSIM is presented in the subsection of the ablation studies in the experiment section (located in Section IV-C4).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

E. Loss Function

We supervise all five-layer outputs of the decoder using a hybrid loss function, including binary cross-entropy (BCE) loss and intersection over union (IOU) loss, i.e.,

$$L = \sum_{i=1}^{5} (l_{\mathrm{BCE}}(S_i, G) + l_{\mathrm{IOU}}(S_i, G)) \tag{20}$$

where S_i is the saliency map output by the decoder of the ith layer of the model, G is GT, and L is the overall loss of the model.

IV. Experiment

A. Experimental Protocol

1) Datasets: We test our model on three classic datasets.

1) *ORSSD [9]:* contains 600 training samples and 200 testing samples. This dataset is the earliest published dataset in the RSI-SOD field. The optical RSI images in this dataset are collected from Google Earth and some existing RSI datasets, and all images are with pixel-level annotations.
2) *EORSSD [10]:* contains 1400 training samples and 600 testing samples, which is an extended version of ORSSD. This dataset contains more diverse samples than ORSSD, and the complexity of the samples is also higher.
3) *ORSI4199 [12]:* contains 2000 training samples and 2199 testing samples, which is the latest released dataset in the field of RSI-SOD. Due to the high diversity and complexity of samples, this dataset is currently the most challenging dataset in the field of RSI-SOD.

2) Evaluation Metrics: To validate the performance of our model, we selected the mainstream evaluation indicators [46]. Suppose the saliency map output by the model is S, and the corresponding GT is G.

1) *PR Curve [47]:* We used 0–255 as the threshold values. Calculate precision and recall under different threshold values, and then drew the PR curve.
2) *S-Measure [48]:* S_α applies object-aware (S_o) and region-aware (S_r) structural similarity to evaluate the similarity of the saliency map to the GT. To meet the mainstream evaluation standards, we put $\alpha = 0.5$.
3) *F-Measure [47]:* F_β comprehensively considers the impact of Precision and Recall. To meet the mainstream evaluation standards, we put $\beta = 0.3$.
4) *E-Measure [49]:* E_ξ jointly considers global image-level statistics and local pixel-level matching information.
5) *MAE [50]:* Mean absolute error (MAE) is the one-norm of the difference between S and G, which can better measure the gap between the generated saliency map and the GT.

3) Implementation Details: All our experiments are conducted on NVIDIA GeForce RTX3090 GPU. The experimental environment is python3.8 under Ubuntu18.04, with Pytorch1.12.1 + cu116 installed.

For the training set, we adopted the method recommended by [9] for data augmentation. Each image in the training set is flipped and rotated to obtain seven additional images. Eventually, the number of training set samples becomes eight times as large.

During training, we resized the input image and the GT to 224 × 224, and randomly flipped the input image, randomly rotated (with continuous values ranging from −90° to 90°), Gaussian blur, and normalized it as well. For the model version of VGG, we set the batch size to 8. For the model version of ResNet, we adjusted the batch size to 20. (Since the first layer of ResNet reduces the H and W dimensions of the input image by half, ResNet versions have smaller memory usage during training compared to VGG.) Both versions of the model are trained for 60 epochs on the training set.

B. Comparison With State-of-the-Art Methods

1) Comparison Methods: We compared our method with 14 state-of-the-art NSI-SOD methods and RSI-SOD methods on three challenging datasets, as illustrated in Table I, including two traditional NSI-SOD methods (regularized random walks ranking (RRWR) [27] and reversion correction and RRWR (RCRR) [25]), three CNN-based NSI-SOD methods (edge guidance network (EGNet) [5], multiscale interactive network (MINet) [4], and GatedNet [23]), two hybrid-based NSI-SOD methods (relational reasoning network with parallel multiscale attention (RRNet) [14] and hybrid feature aligned network (HFANet) [17]), and seven CNN-based RSI-SOD methods (LVNet [9], DAFNet [10], MJRBM [12], EMFINet [11], lightweight salient object detection in optical remote sensing images via feature correlation (CorrNet) [51], ACCoNet [15], and ERPNet [16]).

All the results are produced by running the released source codes with default parameters or acquired from the publicly available results provided by the authors. For the method without code access, we cannot measure their real performance on the ORSI dataset, so they are marked as—in Table I.

2) Quantitative Comparison: We quantitatively compared our method with 14 existing state-of-the-art methods on three datasets, as listed in Table I.

Overall, there are 12 performance metrics on three datasets. Ours-V achieved the best performance on six metrics, while Ours-R achieved the best performance on seven metrics. In addition, all metrics of our model ranked in the top three.

For ORSSD, Ours-V achieved the best results on all four metrics, while Ours-R achieved the best results on three metrics. This shows that our model performs well on small datasets. It is worth noting that Ours-R is outperformed by Ours-V. We argued that the ORSSD dataset is relatively simple, and preventing overfitting is more helpful for performance improvement than improving the model's fitting ability.

For EORSSD, Ours-V achieved the best results on two metrics, while Ours-R achieved the best results on one metric. It can be observed that the F_β of Ours-V is significantly better than Ours-R. It may be due to the fact that ResNet as a backbone reduces the feature map to a smaller size too early. Specifically, the size of the feature map output by the first block of VGG is 224, while ResNet is only 112. The larger the size being processed, the more advantageous it is for improving F_β. HFANet-R surpassed our model in S_α and F_β. This is partly because the input size of HFANet is 448 × 448,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISON OF OUR METHOD WITH 14 METHODS PROPOSED BY OTHER RESEARCHERS. AMONG THE 14 METHODS COMPARED, THERE ARE 2 TRADITIONAL NSI-SOD METHODS (REFERRED TO AS T-NSI), 3 CNN-BASED NSI-SOD METHODS (REFERRED TO AS C-NSI), 2 HYBRID-BASED RSI-SOD METHODS (REFERRED TO AS H-RSI), AND 7 CNN-BASED RSI-SOD METHODS (REFERRED TO AS C-RSI). THE -V AND -R FOLLOWING THE MODEL INDICATE WHETHER THE BACKBONE OF THE MODEL IS VGG OR RESNET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Methods	Publication	Type	FPS	ORSSD				EORSSD				ORSI4199			
				$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$MAE \downarrow$
RRWR [27]	2015 CVPR	T-NSI	0.3	0.6835	0.5590	0.7649	0.1324	0.5992	0.3993	0.6894	0.1677	0.6416	0.5407	0.7116	0.1717
RCRR [25]	2018 TIP	T-NSI	0.3	0.6849	0.5591	0.7651	0.1277	0.6007	0.3995	0.6882	0.1644	0.6491	0.5480	0.7192	0.1637
EGNet [5]	2019 ICCV	C-NSI	-	0.8721	0.8332	0.9731	0.0216	0.8601	0.7880	0.9570	0.0110	0.8516	0.8371	0.9241	0.0385
MINet [4]	2020 CVPR	C-NSI	12	0.9040	0.8761	0.9545	0.0144	0.9040	0.8344	0.9442	0.0093	0.8116	0.7988	0.8961	0.0504
GatedNet [23]	2020 ECCV	C-NSI	25	0.9186	0.8871	0.9664	0.0137	0.9114	0.8566	0.9610	0.0095	0.8545	0.8450	0.9256	0.0393
LVNet [9]	2019 TGRS	C-RSI	1.4	0.8815	0.8263	0.9456	0.0207	0.8630	0.7794	0.9254	0.0146	-	-	-	-
DAFNet-V [10]	2021 TIP	C-RSI	26	0.9191	0.8928	0.9771	0.0113	0.9166	0.8614	0.9861	0.0060	0.8492	0.8348	0.9181	0.0422
MJRBM-V [12]	2022 TGRS	C-RSI	32	0.9204	0.8842	0.9623	0.0163	0.9197	0.8656	0.9646	0.0099	0.8593	0.8493	0.9311	0.0374
EMFINet-V [11]	2022 TGRS	C-RSI	25	0.9366	0.9002	0.9737	0.0109	0.9290	0.8720	0.9711	0.0084	0.8692	0.8608	0.9373	0.0327
CorrNet-V [51]	2022 TGRS	C-RSI	100	0.9380	0.9129	0.9790	0.0098	0.9289	0.8778	0.9696	0.0083	0.8626	0.8560	0.9333	0.0366
ACCoNet-V [15]	2023 TCYB	C-RSI	81	0.9437	0.9149	0.9796	0.0088	0.9290	0.8837	0.9727	0.0074	0.8775	0.8687	0.9412	0.0314
ERPNet-V [16]	2023 TCYB	C-RSI	50	0.9254	0.8974	0.9710	0.0135	0.9210	0.8632	0.9603	0.0089	0.8642	0.8551	0.9284	0.0376
Ours-V	-	C-RSI	25	**0.9460**	**0.9183**	**0.9828**	**0.0086**	**0.9358**	**0.8838**	**0.9751**	**0.0079**	**0.8702**	**0.8670**	**0.9403**	**0.0325**
MJRBM-R [12]	2022 TGRS	C-RSI	22	0.9211	0.8885	0.9686	0.0145	0.9091	0.8555	0.9655	0.0099	0.8582	0.8511	0.9343	0.0372
EMFINet-R [11]	2022 TGRS	C-RSI	25	0.9432	0.9155	0.9813	0.0095	0.9319	0.8742	0.9712	0.0075	0.8712	0.8636	0.9403	0.0313
RRNet-R [14]	2022 TGRS	H-RSI	109	0.9339	0.9011	0.9722	0.0113	0.9266	0.8743	0.9665	0.0082	0.8585	0.8500	0.9286	0.0367
HFANet-R [17]	2022 TGRS	H-RSI	26	0.9399	0.9117	0.9770	0.0092	0.9380	0.8876	0.9740	0.0071	0.8767	0.8700	0.9431	0.0314
ACCoNet-R [15]	2023 TCYB	C-RSI	-	0.9428	0.9149	0.9819	0.0087	0.9302	0.8821	0.9759	0.0067	0.8805	0.8688	0.9424	0.0320
ERPNet-R [16]	2023 TCYB	C-RSI	50	0.9352	0.9036	0.9738	0.0114	0.9252	0.8743	0.9665	0.0082	0.8636	0.8528	0.9292	0.0388
Ours-R	-	C-RSI	23	**0.9455**	**0.9160**	**0.9814**	**0.0085**	**0.9347**	**0.8792**	**0.9757**	**0.0064**	**0.8755**	**0.8726**	**0.9459**	**0.0305**

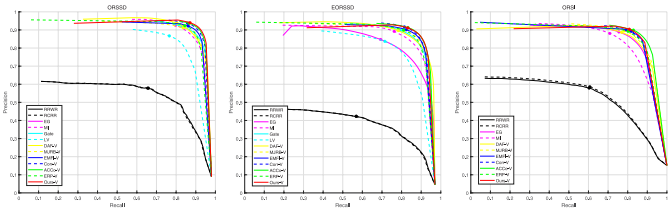


Fig. 8. Comparison of PR curves of Ours with other state-of-the-art methods. The closer the curve is to the upper right corner, the better the performance of the method. The PR curves on both datasets demonstrate the superiority of our proposed method.

twice the size of Ours-V, and it uses a CNN + transformer hybrid encoder.

For ORSI4199, Ours-V achieved the second-best results on four metrics, while Ours-R achieved the best results on three metrics. Although ACCoNet-V surpassed our model, the gap between our model's F_β and E_ξ and ACCoNet-V is not large, and Ours-V significantly surpassed ACCoNet-R on three metrics.

Additionally, we compared the PR curves of different methods on three datasets. As shown in Fig. 8, the solid red curves are the PR curve of Ours. The PR curves of our method on ORSSD and EORSSD are the closest to the upper right corner, which also proves that our method is the optimal method.

3) Qualitative Comparison: To qualitatively compare our method with other state-of-the-art methods, we selected a subset of the most challenging test samples in EORSSD. According to the different challenges, we divided these samples into five parts. The comparisons between the saliency maps of our method and other state-of-the-art methods are shown in Fig. 9.

From Fig. 9 we can observe that, Part 1 lists some images with heavy shadows. Optical remote sensing images are bird's-eye views taken from high altitudes, where shadows are often interlaced with salient objects. This brings great difficulty to the detection of salient objects. Methods such as ERPNet and LVNet all falsely detect shadows as part of the salient objects. Compared with other state-of-the-art methods, our method can better avoid the misleading of shadows, which is attributed to our proposed DSIM and MFEM. For complex patch shadows as shown in the first row, MFEM can identify salient objects from multiple scales and remove the negative effects of shadows. And for the large shadow blocks shown in the second row, DSIM can exclude them from a semantic point of view. Therefore, for shadows, our method obtains better performance.

Part 2 presents images with multiple salient objects. The possibility of having multiple salient objects is a characteristic of optical remote sensing images, making the saliency detection task difficult. It can be seen from the figure that methods such as ERPNet, EMFINet, and MJRBM cannot detect all the salient objects in the image. However, our method effectively detects all salient objects in the image. This is because our proposed MFEM can accurately localize multiple salient objects from multiple scales.

Part 3 shows images of salient objects with complex edges, which are often found in salient objects such as islands and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

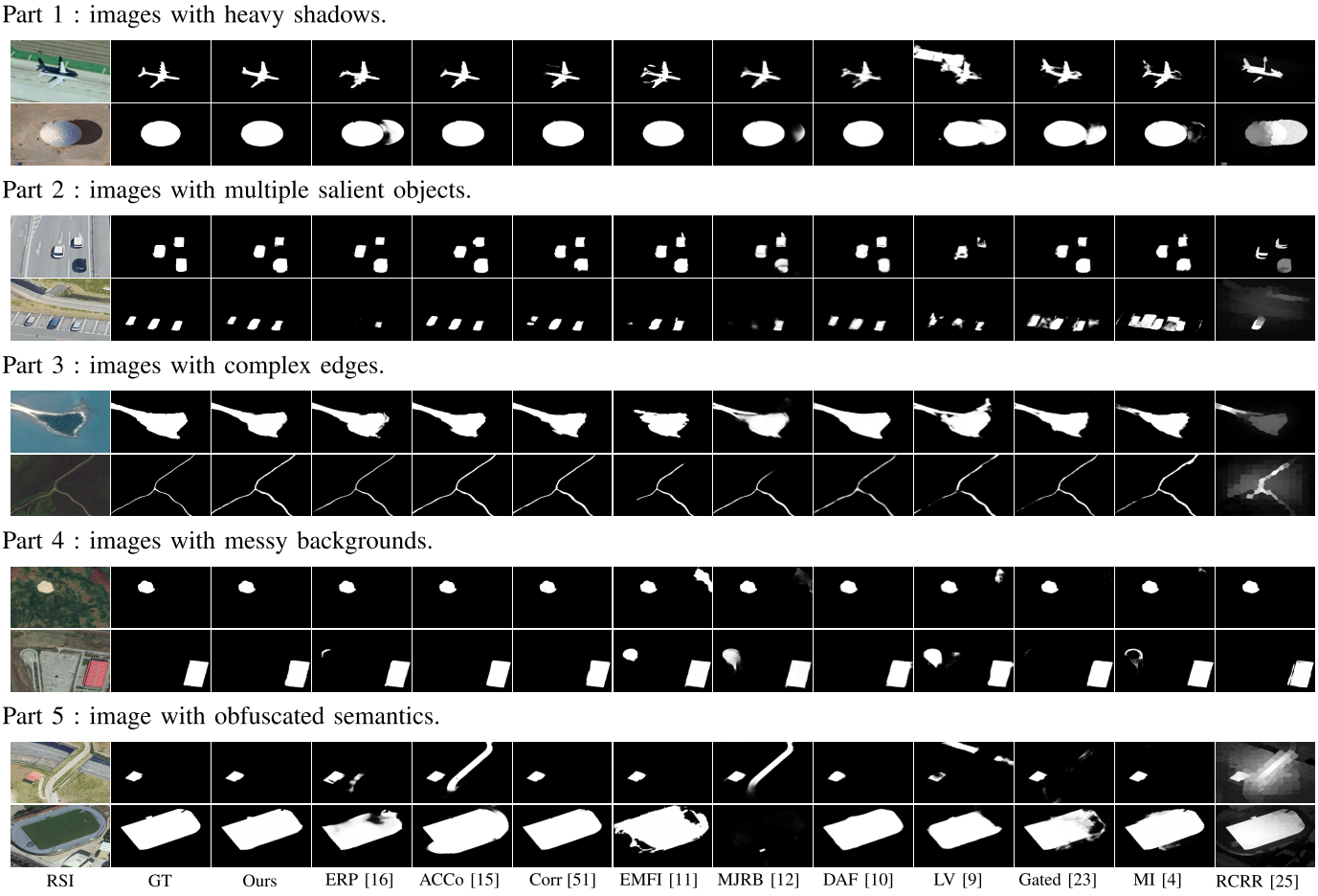


Fig. 9. Comparison of ours-V with other state-of-the-art models (−V) on different challenging parts of the EORSSD testing data.

rivers in optical remote sensing images. Methods including EMFI, MJRB, DAFNet, and LVNet cannot accurately capture the edge details. In contrast, our model can capture finer edge information. This can be attributed to LDAM, which can adaptively perceive edge features of salient objects without edge guidance.

Part 4 presents images with messy backgrounds, which greatly disturbs the accuracy of SOD. Methods such as EMFINet, MJRBM, and LVNet all falsely detect a part of the background as salient objects. However, our method does not suffer from false detections. Our proposed DSIM can analyze the semantic features of images deeply, thereby avoiding the problem of detecting complex backgrounds as salient objects.

Part 5 presents images with obfuscated semantics, which refers to objects with multiple different semantics in an image, such as roads, squares, and houses appearing in one image at the same time. When this happens, the model is supposed to be able to detect the most salient objects. Compared with other models, our method can avoid the interference of unimportant objects and accurately identify salient objects. This mainly relies on our proposed DSIM. Through deep semantic interaction, the model tends to detect the targeted objects.

4) Attribute Analysis on ORSI4199: In the challenging ORSI-4199 dataset, each image has attributes that reflect typical challenges in remote sensing scenes. The specific meaning of each attribute can refer to [12]. Table II shows the F_β of our proposed model and other state-of-the-art models. The results show that our model ranks first in 5 out of 9 attributes. In addition, our model's average performance ranks first. These results indicate that our proposed model outperforms the existing models in most challenging situations.

TABLE II

ATTRIBUTE ANALYSIS ON THE ORSI4199 DATASET. TO MEET THE MAINSTREAM EVALUATION STANDARDS, ALL THE MODELS INVOLVED IN THE COMPARISON ARE RESNET VERSIONS EXCEPT THAT CORRNET ONLY PROVIDES THE VGG VERSION. THE TOP THREE MAX F_β ARE SHOWN IN RED, BLUE AND GREEN

Attr	MJRBM [12]	RRNet [14]	CorrNet [51]	HFANet [17]	ERPNet [16]	ACCoNet [15]	Ours
BSO	0.9166	0.8811	0.8891	0.9103	0.8946	0.9091	0.9109
CS	0.8961	0.8702	0.8835	0.8964	0.8800	0.8974	0.9068
CSO	0.8899	0.8649	0.8728	0.8948	0.8727	0.8931	0.9001
ISO	0.8937	0.8594	0.8742	0.9046	0.8837	0.8997	0.9093
LCO	0.7916	0.7710	0.7852	0.8030	0.7925	0.8048	0.8136
MSO	0.8585	0.8288	0.8528	0.8566	0.8468	0.8611	0.8579
NSO	0.8748	0.8392	0.8724	0.8858	0.8467	0.8814	0.8922
OC	0.8422	0.8042	0.8409	0.8506	0.8378	0.8337	0.8349
SSO	0.8091	0.7682	0.8051	0.8128	0.8015	0.8065	0.8079
Avg	0.8636	0.8319	0.8529	0.8682	0.8507	0.8652	0.8704

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION EXPERIMENTS PERFORMED ON ORSSD AND EORSSD. WHENEVER WE REMOVE A MODULE, THE MODEL PERFORMANCE ALWAYS DEGRADES. THIS VERIFIES THE EFFECTIVENESS OF EACH OF OUR PROPOSED MODULES. THE BEST OF THESE RESULTS ARE SHOWN IN BOLD

No.	LDAM	MFEM	DSIM	ORSSD			EORSSD		
				S_α	F_β	E_ξ	S_α	F_β	E_ξ
1				0.9278	0.8887	0.9629	0.9220	0.8628	0.9602
2	✓			0.9364	0.9062	0.9735	0.9295	0.8721	0.9652
3	✓	✓		0.9436	0.9142	0.9804	0.9331	0.8781	0.9710
4	✓	✓	✓	**0.9460**	**0.9183**	**0.9828**	**0.9358**	**0.8838**	**0.9751**

Our model significantly outperforms other models under complex scene (CS) and low contrast scene (LCO) conditions, demonstrating its ability to identify salient objects from cluttered backgrounds. Under complex salient object (CSO) and narrow salient object (NSO) conditions, our model also shows advantages, proving its strong edge extraction ability. Under small salient object (SSO), big salient object (BSO), and multiple salient objects (MSO) conditions, although our model's score did not reach the highest, the gap with the first-ranked score was insignificant, proving that our model has good multiscale feature extraction capabilities. However, the performance of our model under off center (OC) conditions still needs to be improved.

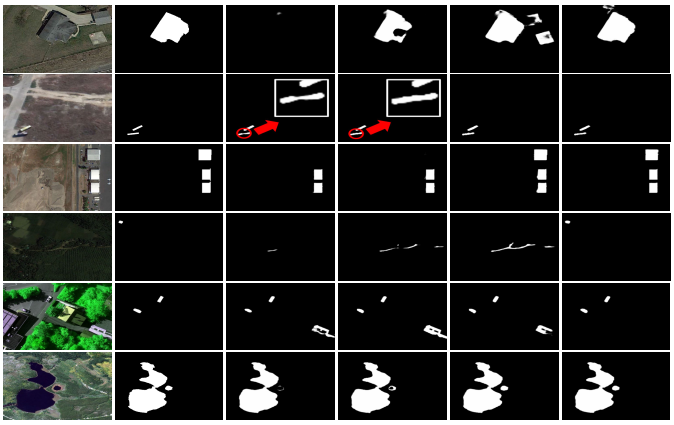
C. Ablation Study

In this section, we conducted comprehensive ablation experiments on the proposed model to verify the effectiveness of our proposed method.

1) Ablation Experiments Between Different Modules: We conducted ablation experiments on ORSSD and EORSSD datasets to verify the effectiveness of each module in the model. For all ablation experiments, we ensured that the hyperparameters used and the running environment remain unchanged.

a) Baseline: We adopted VGG16 as the encoder of the baseline, and used the decoder mentioned in the model architecture diagram (the output of the deep decoder is dimensionally concatenated with the output of the encoder in the same layer, and two layers of Conv $3 \times 3+$ BN $+$ ReLU operations are performed). We applied BCE and IOU loss on all five-layer decoder outputs for supervision, and took the output of the shallowest decoder as the final result of the model. As shown in No.1 in Table III, on ORSSD, the performance of the baseline is 0.9278 for S_α, 0.8887 for F_β, and 0.9629 for E_ξ. On EORSSD, the performance of the baseline is 0.9220 for S_α, 0.8628 for F_β, and 0.9602 for E_ξ.

b) Effects of LDAM: Based on the baseline, we applied the LDAM to the output of the first, second, and third layers of the encoder. It should be noted that since the deep layer of the LDAM in the second layer has no other enhancer modules, the interaction between the layers is only passed from the LDAM of the second layer to the LDAM of the first layer. As shown in No.2 in Table III, the overall performance of the model has been improved to a certain extent by applying LDAM. To be specific, on EORSSD, the improvements of S_α and F_β are relatively large, which are respectively 0.75% and 0.93% higher than the baseline. As shown in the second row of Fig. 10, compared with the baseline [i.e., column 10(c)], the edge of salient objects predicted by the model added with LDAM [i.e., column 10(d)] is closer to the GT.



(a) (b) (c) (d) (e) (f)

Fig. 10. Visual examples of ablation experiments between different modules on EORSSD. (a) Optical RSIs. (b) GT. (c) Baseline. (d) Baseline + LDAM. (e) Baseline + LDAM + MFEM. (f) Baseline + LDAM + MFEM + DSIM (ours).

c) Effects of MFEM: Based on No.2, we applied the MFEM to the outputs of the fourth and fifth layers of the encoder. It should be noted that the MFEM of the fifth layer does not accept the guidance from the upper layer, while the MFEM of the fourth layer passes the output as the upper layer guidance to the LDAM of the third layer. The test results are shown in No.3 in Table III. It can be found that compared with No.2, on EORSSD, the S_α of this model is increased by 0.36%, the F_β is by 0.60%, and the E_ξ is by 0.58%. As shown in the third row of Fig. 10, after adding MFEM [i.e., column 10(e) and (f)], the model can accurately identify multiscale salient objects.

d) Effects of DSIM: We added the DSIM on the basis of model No.3. The DSIM module aggregates the sideouts of the deepest three layers of the encoder, extracts global contextual information through multiscale graph convolution, and applies it to the fifth-layer MFEM as high-level guidance. After adding DSIM, the performance of the model is shown in No.4 in Table III. On EORSSD, the contribution of the DSIM to the overall model is mainly reflected in F_β and E_ξ, which is increased by 0.57% and 0.41%. As shown in the fourth row of Fig. 10, after adding DSIM [i.e., column 10(f)], the model can effectively avoid the interference of the cluttered background and accurately detect salient objects.

From the visual effects shown in Fig. 10, the effectiveness of the proposed LDAM, MFEM, and DSIM has also been verified. For the second row, when LDAM is added, the edges of salient objects are closer to the GT. For the third row, when MFEM is added, the missing salient objects are detected. This proves that MFEM has good adaptability to multiobject scenes. For the sixth row, when MFEM is added, the model can better adapt to salient objects with large-scale changes and accurately identify large- and small-scale salient objects. For the first row, the Baseline did not detect any salient objects and the version without DSIM had false positives and false negatives. For the fourth and fifth rows, when DSIM is added, the model corrects errors and correctly detects salient objects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
TEST RESULTS OF THE LDAM INTERNAL ABLATION EXPERIMENTS SHOW THAT THE DIFFERENCE PERCEPTION PLAYS A MAJOR CONTRIBUTION TO PERFORMANCE IMPROVEMENT. THE BEST OF THESE RESULTS ARE SHOWN IN BOLD

Version	ORSSD			EORSSD		
	S_α	F_β	E_ξ	S_α	F_β	E_ξ
Ours-V	**0.9460**	**0.9183**	**0.9828**	**0.9358**	**0.8838**	**0.9751**
$w/o\ DP$	0.9430	0.9126	0.9725	0.9332	0.8789	0.9723
$w/o\ DP\&UB$	0.9451	0.9147	0.9820	0.9330	0.8769	0.9707
$w/o\ DP\&LB$	0.9450	0.9174	0.9824	0.9342	0.8798	0.9730

TABLE V
TEST RESULTS OF THE MFEM INTERNAL ABLATION EXPERIMENTS. THIS EXPERIMENT DEMONSTRATES THE EFFECTIVENESS OF THE MULTISCALE DILATED CONVOLUTION IN MFEM. THE BEST OF THESE RESULTS ARE SHOWN IN BOLD

Version	ORSSD			EORSSD		
	S_α	F_β	E_ξ	S_α	F_β	E_ξ
Ours-V	**0.9460**	**0.9183**	**0.9828**	**0.9358**	**0.8838**	**0.9751**
$w/o\ UB$	0.9459	0.9171	0.9823	0.9347	0.8812	0.9725
$w/o\ DB$	0.9456	0.9153	0.9815	0.9348	0.8799	0.9713
$w/\ NC$	0.9458	0.9148	0.9800	0.9331	0.8775	0.9702
$r/\ C$	0.9439	0.9128	0.9795	0.9329	0.8780	0.9702

This indicates that DSIM can infer whether a block contains a salient object from a semantic feature perspective and guide shallow feature extraction in the form of attention maps.

All in all, we verified the correctness of each module through the above ablation experiments. We find that each time a module is added, the overall performance of the model will be improved, which shows that all our proposed modules are effective.

2) Ablation Experiment Inside the LDAM: To verify the effectiveness of the internal structure of LDAM, we conducted ablation experiments for the framework of LDAM. We built three mutilated versions of LDAM as follows.

1) $w/o\ DP$*:* LDAM without difference perception.
2) $w/o\ DP\&UB$*:* LDAM without difference perception and upper branch.
3) $w/o\ DP\&LB$*:* LDAM without difference perception and lower branch.

We applied the three versions of LDAM to the model, tested the performance results of the three models on EORSSD, and compared them with the results of the original model. The inference speed of the above three variants is almost the same as that of our proposed AESINet (25FPS). The test results of the three models are shown in Table IV.

It can be found that after removing the difference perception mechanism of the model, the performance of the model has declined to a certain extent. This demonstrates the effectiveness of our proposed difference perception mechanism. When the difference perception mechanism is removed, the feature maps of the two branches will not be able to perceive the difference information between each other, and the model will lose the ability to enhance the edge features to a certain extent.

When the difference perception mechanism is removed along with a branch, there is a slight drop in performance on EORSSD compared to only removing the difference perception mechanism. This shows that the double-branched convolutional structure plays a secondary role in improving the effect of the LDAM. As for the abnormal rise phenomenon on ORSSD, we believe that the performance of LDAM is not fully exerted due to the relatively low challenge of ORSSD and less complex edges. Therefore, we believe that the difference perception mechanism is the core of LDAM.

As shown in part (a) of Fig. 11, the edges of salient objects predicted by the model applying the three variant versions of LDAM are relatively blurred, while the original AESINet (Ours) performs better in terms of edge accuracy. This also proves the effectiveness of LDAM.

3) Ablation Experiment Inside the MFEM: To verify the effectiveness of multiscale dilated convolution in MFEM, we designed ablation experiments on the internal structure of the MFEM. Specifically, we verified the effectiveness of scaling the feature maps to three different sizes and the effectiveness of applying dilated convolutions with different dilation rates to the feature maps. We designed four variants of MFEM as follows.

1) $w/o\ UB$*:* MFEM without the branch that enlarged feature maps.
2) $w/o\ DB$*:* MFEM without the branch that downscaled feature maps.
3) $w/\ NC$*:* MFEM replacing dilated convolutions with normal convolutions.
4) $r/\ C$*:* Replace the convolution of the three branches in MFEM with traditional parallel dilated convolution (keep only the middle branch).

We tested the performance of the above three variants of MFEM, and the test results are shown in Table V. The inference speed of the above four variants is almost the same as that of our proposed AESINet (25FPS).

For 1) and 2), we found that when one of the dilated convolution branches of three sizes was removed, there was a small decrease in model performance. We argued that this is mainly due to the fact that the variant MFEM removes branches of a certain size, and the multiscale feature extraction ability has slightly decreased.

For 3), when we replaced all the multiple parallel convolutions with ordinary convolutions, the model performance also decreased to a certain extent. We can infer that multiple parallel convolutions have the effect of increasing the receptive field in the model and can more precisely locate the positions of salient objects.

For 4), when we replaced the three branches of MFEM with traditional parallel dilated convolution, the performance of the model decreased to some extent. This is because traditional parallel dilated convolution only changes the spacing among the elements of each convolution kernel and does not change the coverage of each convolution kernel element. In contrast, our MFEM scales the feature map to three different scales, indirectly allowing the convolution kernel elements to cover different sizes of ranges, enhancing the multiscale feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

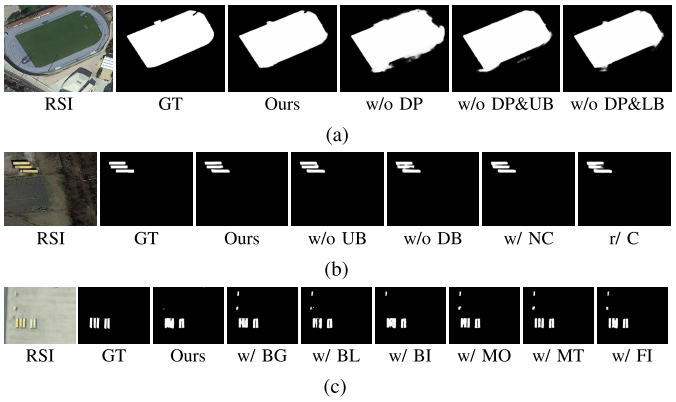


Fig. 11. Visual examples of ablation experiments inside modules on EORSSD. (a) Inside LDAM. (b) Inside MFEM. (c) Inside DSIM.

TABLE VI

RESULTS OF ABLATION EXPERIMENTS INSIDE DSIM. THIS EXPERIMENT VERIFIES THE EFFECTIVENESS OF THREE PARALLEL GRAPH CONVOLUTION BRANCHES IN DSIM. THE EXPERIMENT ALSO VERIFIED THE RATIONALITY OF MAPPING EACH PIXEL TO TWO GRAPH VERTICES. THE BEST OF THESE RESULTS ARE SHOWN IN BOLD

Version	ORSSD			EORSSD		
	S_α	F_β	E_ξ	S_α	F_β	E_ξ
Ours-V	**0.9460**	0.9183	**0.9828**	**0.9358**	**0.8838**	0.9751
w/ BG	0.9447	0.9167	0.9823	0.9348	0.8801	0.9728
w/ BL	0.9445	**0.9184**	0.9823	0.9334	0.8789	0.9701
w/ BI	0.9451	0.9165	0.9826	0.9343	0.8820	0.9741
w/ MO	0.9447	0.9124	0.9794	0.9335	0.8794	0.9711
w/ MT	0.9431	0.9154	0.9814	0.9333	0.8792	**0.9764**
w/ FI	0.9458	0.9161	0.9813	0.9343	0.8828	0.9735

extraction capability. Therefore, it makes sense for us to scale the feature maps to three different sizes.

As shown in part (b) of Fig. 11, the model applying the three variant versions of MFEM performed poorly in detecting multiple salient objects, while the original AESINet (Ours) accurately detected multiple salient objects. Therefore, we have reason to believe that the structures in MFEM are all valid.

4) Ablation Experiment Inside the DSIM: To verify the effectiveness of the structure of DSIM, we conducted ablation experiments on the internal structure of DSIM. Specifically, we verified the effectiveness of applying 1×1 convolutions to the concatenated inputs for dimensionality reduction, and the effectiveness of graph convolution interactions for three parallel paths. We built six variants of DSIM as follows.

1) *w/ BG:* DSIM that only builds graph edges within all 7×7 pixel grid.
2) *w/ BL:* DSIM that only builds graph edges within 2×2 pixel grid.
3) *w/ BI:* DSIM that only builds graph edges within one pixel grid.
4) *w/ MO:* DSIM that each pixel grid maps to one graph vertex.
5) *w/ MT:* DSIM that each pixel grid maps to three graph vertices.
6) *w/ FI:* The input of the DSIM is changed to all five outputs of the encoder.

As shown in Table VI, when changing the structure in DSIM, the model performance degrades to some extent. Regarding the speed, except that the inference speed of variant 4) is 33FPS and 2) is 20FPS, the inference speed of other variants is almost the same as that of our proposed AESINet (25FPS). This shows that the internal structures of our proposed DSIM modules are all effective.

For 1), 2) and 3), when we only keep a certain edge construction method, the performance of the model decreases on both ORSSD and EORSSD. This is because global interaction, local interaction, and self-interaction greatly enhance the semantic recognition ability of DSIM. When the branches of global interaction, local interaction, or self-interaction are removed, it is difficult for the model to make an effective semantic analysis of salient objects in small regions. As shown in part (c) of Fig. 11, the models using the three variant versions of DSIM all mistakenly recognized some parts of the background as salient objects, while the original AESINet (Ours) effectively avoided the interference caused by the cluttered background. Through the above analysis, the three parallel graph convolution branches (i.e., global interaction, local interaction, and self-interaction) all benefit the semantic reasoning of salient objects.

For 4), when the number of vertices is specified as one, the deep semantic reasoning ability of DSIM is decreased, manifested as a drop in the overall performance of the model. This shows that only one graph node is not enough to characterize complex semantic features of a region. For 5), when the number of vertices is specified as three, although the E_ξ of EORSSD surpasses the original model, the other five metrics are lower than the original model. At the same time, mapping to three graph vertices is computationally more expensive. In summary, it is reasonable for us to set each pixel to be mapped to two graph vertices.

For 6), when we chose to aggregate all the outputs of the encoder, the shallow outputs of the encoder have a negative effect on the reasoning of the deep semantic information, which reduces the performance of the model. We can analyze according to the properties of DSIM. When performing graph inference, DSIM uses a 7×7 feature map, where each pixel can be mapped to two graph nodes. In fact, each pixel on the 7×7 feature map represents rich semantic information about the region. Deeper encoder blocks can better abstract the semantic features of images. The output of the shallower encoder block represents more detailed features, such as edges and textures. Therefore, we preserved the deep abstract semantic features while removing the shallow texture detail features.

5) Validation of Network Architecture Validity: To provide more intuitive and powerful evidence to support the rationality of our network structure, we conducted experiments on the network architecture. The variants of AESINet regarding network architecture are shown below.

1) *rLwM:* Replace all LDAM with MFEM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
RESULTS UNDER DIFFERENT NETWORK STRUCTURES. THIS EXPERIMENT PROVES THAT IT IS REASONABLE FOR US TO APPLY LDAM AT LAYERS 1, 2, AND 3 AND MFEM AT LAYERS 4 AND 5. THE BEST OF THESE RESULTS ARE SHOWN IN BOLD

Version	ORSSD			EORSSD		
	S_α	F_β	E_ξ	S_α	F_β	E_ξ
Ours-V	**0.9460**	**0.9183**	**0.9828**	**0.9358**	**0.8838**	**0.9751**
$rLwM$	0.9439	0.9128	0.9795	0.9347	0.8814	0.9731
$rMwL$	0.9445	0.9127	0.9799	0.9346	0.8813	0.9727
sLM	0.9442	0.9113	0.9772	0.9329	0.8785	0.9704
$L2M3$	0.9458	0.9175	0.9821	**0.9358**	0.8835	0.9741

2) $rMwL$: Replace all MFEM with LDAM.
3) sLM: Swap the positions of LDAM and MFEM.
4) $L2M3$: Apply LDAM to the first and second layers and apply MFEM to the third, fourth, and fifth layers.

As shown in Table VII, the performance of our original version of AESINet surpasses several variant versions. The specific reason can be understood according to the characteristics of the feature map. For the encoder, the attributes of the feature maps output by different layers are different. In the shallow layer, the feature map output by the Encoder has rich edge and texture features. As the number of layers deepens, these detailed features are gradually abstracted into semantic features. LDAM is suitable for enhancing edges with detailed features, and its application to shallow features yields highly favorable outcomes. Applying MFEM to deep features aims to better recognize multiscale objects at the level of abstract features.

For variant 1), LDAM was replaced with MFEM, and the model performance declined. This is because MFEM does not perform well in detecting shallow detail information. In addition, compared to LDAM, MFEM lacks the ability to enhance edges. For variants 2) and 3), similarly, the model performance also declined. This shows that our proposed LDAM is indeed suitable for dealing with shallow fine features, while MFEM is suitable for further mining its multiscale connotation information from the deep abstract feature level.

Regarding the question of why LDAM is not applied to the first and second layers and why MFEM is not applied to the third, fourth, and fifth layers. We believe this is an empirical setting. In fact, we think it is reasonable to apply either LDAM or MFEM to the third layer. As shown in variant d), its performance is not much different from the AESINet we proposed. The performance of applying LDAM to the third layer is slightly higher than that of applying MFEM to the third layer. Therefore, we chose to apply LDAM in the first three layers and MFEM in the last two layers.

V. CONCLUSION

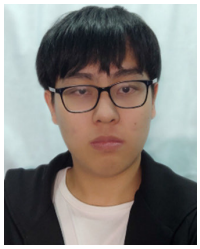
In this article, we focus on the application of SOD in the field of optical RSI and propose AESINet. The model is equipped with LDAM, MFEM, and DSIM. LDAM captures intricate details, such as edges, through the difference perception between two-stream convolutions and then sharpens the edges of salient objects. MFEM effectively extracts multiscale information of salient objects by applying multiple parallel dilated convolutions at three different scales. DSIM performs convolution operations on three differently constructed graphs, enabling effective interaction of deep semantic information. By leveraging these methods, the proposed AESINet can accurately detect salient objects in optical RSI. Extensive experiments are performed on three challenging datasets, and the results demonstrate the superiority of AESINet.

REFERENCES

[1] A. Borji, M.-M. Cheng, H. Jiang, and J. Li, "Salient object detection: A benchmark," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5706–5722, Dec. 2015.
[2] R. Cong, J. Lei, H. Fu, M.-M. Cheng, W. Lin, and Q. Huang, "Review of visual saliency detection with comprehensive information," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 10, pp. 2941–2959, Oct. 2019.
[3] W. Wang, Q. Lai, H. Fu, J. Shen, H. Ling, and R. Yang, "Salient object detection in the deep learning era: An in-depth survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 3239–3259, Jun. 2022.
[4] Y. Pang, X. Zhao, L. Zhang, and H. Lu, "Multi-scale interactive network for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9410–9419.
[5] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF ICCV*, Oct. 2019, pp. 8778–8787.
[6] T. Zhao and X. Wu, "Pyramid feature attention network for saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3080–3089.
[7] Q. Zhai, X. Li, F. Yang, C. Chen, H. Cheng, and D.-P. Fan, "Mutual graph learning for camouflaged object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12992–13002.
[8] M. Xu, P. Fu, B. Liu, H. Yin, and J. Li, "A novel dynamic graph evolution network for salient object detection," *Int. J. Speech Technol.*, vol. 52, no. 3, pp. 2854–2871, Feb. 2022.
[9] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.
[10] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.
[11] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634819.
[12] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[13] G. Li, Z. Liu, W. Lin, and H. Ling, "Multi-content complementation network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614513.
[14] R. Cong, Y. Zhang, L. Fang, J. Li, Y. Zhao, and S. Kwong, "RRNet: Relational reasoning network with parallel multiscale attention for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613311.
[15] G. Li, Z. Liu, D. Zeng, W. Lin, and H. Ling, "Adjacent context coordination network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 526–538, Jan. 2023.
[16] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.
[17] Q. Wang, Y. Liu, Z. Xiong, and Y. Yuan, "Hybrid feature aligned network for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624915.
[18] L. Zhang, A. Li, Z. Zhang, and K. Yang, "Global and local saliency analysis for the extraction of residential areas in high-spatial-resolution remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 3750–3763, Jul. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.

[19] L. Zhang, D. Chen, J. Ma, and J. Zhang, "Remote-sensing image superresolution based on visual saliency analysis and unequal reconstruction networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4099–4115, Jun. 2020.
[20] L. Zhang and J. Zhang, "A novel remote-sensing image fusion method based on hybrid visual saliency analysis," *Int. J. Remote Sens.*, vol. 39, no. 22, pp. 7942–7964, Nov. 2018.
[21] X. Gong, Z. Xie, Y. Liu, X. Shi, and Z. Zheng, "Deep salient feature based anti-noise transfer network for scene classification of remote sensing imagery," *Remote Sens.*, vol. 10, no. 3, p. 410, Mar. 2018.
[22] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.
[23] X. Zhao, Y. Pang, L. Zhang, H. Lu, and L. Zhang, "Suppress and balance: A simple gated network for salient object detection," in *Proc. ECCV*, 2020, pp. 35–51.
[24] Z. Liu, W. Zou, and O. Le Meur, "Saliency tree: A novel saliency detection framework," *IEEE Trans. Image Process.*, vol. 23, no. 5, pp. 1937–1952, May 2014.
[25] Y. Yuan, C. Li, J. Kim, W. Cai, and D. D. Feng, "Reversion correction and regularized random walk ranking for saliency detection," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1311–1322, Mar. 2018.
[26] J.-G. Yu, J. Zhao, J. Tian, and Y. Tan, "Maximal entropy random walk for region-based visual saliency," *IEEE Trans. Cybern.*, vol. 44, no. 9, pp. 1661–1672, Sep. 2014.
[27] C. Li, Y. Yuan, W. Cai, Y. Xia, and D. Dagan Feng, "Robust saliency detection via regularized random walks ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2710–2717.
[28] H. Peng, B. Li, H. Ling, W. Hu, W. Xiong, and S. J. Maybank, "Salient object detection via structured matrix decomposition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 818–832, Apr. 2017.
[29] L. Zhou, Z. Yang, Z. Zhou, and D. Hu, "Salient region detection using diffusion process on a two-layer sparse graph," *IEEE Trans. Image Process.*, vol. 26, no. 12, pp. 5882–5894, Dec. 2017.
[30] Y. Zhou, S. Huo, W. Xiang, C. Hou, and S.-Y. Kung, "Semi-supervised salient object detection using a linear feedback control system model," *IEEE Trans. Cybern.*, vol. 49, no. 4, pp. 1173–1185, Apr. 2019.
[31] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3912–3921.
[32] Z. Chen, Q. Xu, R. Cong, and Q. Huang, "Global context-aware progressive aggregation network for salient object detection," in *Proc. AAAI*, 2020, pp. 10599–10606.
[33] S. Chen, B. Wang, X. Tan, and X. Hu, "Embedding attention and residual network for accurate salient object detection," *IEEE Trans. Cybern.*, vol. 50, no. 5, pp. 2050–2062, May 2020.
[34] S. Chen, X. Tan, B. Wang, and X. Hu, "Reverse attention for salient object detection," in *Proc. ECCV*, 2018, pp. 236–252.
[35] P. Zhang, D. Wang, H. Lu, H. Wang, and X. Ruan, "Amulet: Aggregating multi-level convolutional features for salient object detection," in *Proc. IEEE/CVF ICCV*, Oct. 2017, pp. 202–211.
[36] L. Zhang, J. Dai, H. Lu, Y. He, and G. Wang, "A bi-directional message passing model for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1741–1750.
[37] X. Qin, Z. Zhang, C. Huang, C. Gao, M. Dehghan, and M. Jagersand, "BASNet: Boundary-aware salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7471–7481.
[38] H. Zhou, X. Xie, J.-H. Lai, Z. Chen, and L. Yang, "Interactive two-stream decoder for accurate and fast saliency detection," in *Proc. IEEE/CVF CVPR*, Jun. 2020, pp. 9138–9147.
[39] Q. Hou, M.-M. Cheng, X. Hu, A. Borji, Z. Tu, and P. H. S. Torr, "Deeply supervised salient object detection with short connections," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 4, pp. 815–828, Apr. 2019.
[40] M. Xu, P. Fu, B. Liu, and J. Li, "Multi-stream attention-aware graph convolution network for video salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 4183–4197, 2021.
[41] M. Xu, B. Liu, P. Fu, J. Li, Y. H. Hu, and S. Feng, "Video salient object detection via robust seeds extraction and multi-graphs manifold propagation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 7, pp. 2191–2206, Jul. 2020.
[42] M. Xu, B. Liu, P. Fu, J. Li, and Y. H. Hu, "Video saliency detection via graph clustering with motion energy and spatiotemporal objectness," *IEEE Trans. Multimedia*, vol. 21, no. 11, pp. 2790–2805, Nov. 2019.
[43] B. Liu, K. Mu, M. Xu, F. Wang, and L. Feng, "A novel spatiotemporal attention enhanced discriminative network for video salient object detection," *Int. J. Speech Technol.*, vol. 52, no. 6, pp. 5922–5937, Apr. 2022.
[44] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[45] Y. Li and A. Gupta, "Beyond grids: Learning graph representations for visual recognition," in *Proc. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–12.
[46] D.-P. Fan, Z. Lin, Z. Zhang, M. Zhu, and M.-M. Cheng, "Rethinking RGB-D salient object detection: Models, data sets, and large-scale benchmarks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2075–2089, May 2021.
[47] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.
[48] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE/CVF ICCV*, Oct. 2017, pp. 4558–4567.
[49] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 698–704.
[50] F. Perazzi, P. Krähenbühl, Y. Pritch, and A. Hornung, "Saliency filters: Contrast based filtering for salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 733–740.
[51] G. Li, Z. Liu, Z. Bai, W. Lin, and H. Ling, "Lightweight salient object detection in optical remote sensing images via feature correlation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601111.



Xiangyu Zeng is currently pursuing the B.S. degree with the Department of Artificial Intelligence, School of Computer Science and Technology, Shandong University, Qingdao, China.

He is a Research Assistant with the School of Software, Shandong University, Jinan, China. His research interests include computer vision, pattern recognition, and remote sensing.



Mingzhu Xu (Member, IEEE) received the B.S., M.Sc., and Ph.D. degrees from the Harbin Institute of Technology (HIT), Harbin, China, in 2013, 2015, and 2021, respectively.

He is currently an Assistant Professor with the School of Software, Shandong University, Jinan, China. His research interests include computer vision, multimedia computing, and information retrieval, and explainable artificial intelligence (AI).

Dr. Xu is an invited reviewer for prestigious journals, including IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, *Information Science*, ACM International Conference on Multimedia (ACM MM), International Conference on Machine Learning (ICML), Conference and Workshop on Neural Information Processing Systems (NeurIPS), IEEE INTERNATIONAL CONFERENCE ON ACOUSTICS, SPEECH, AND SIGNAL PROCESSING, IEEE *International Conference on Multimedia and Expo*, and IEEE *International Conference on Pattern Recognition*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.



Yijun Hu is currently pursuing the B.S. degree with the School of Software, Shandong University, Jinan, China.

Her research interests include computer vision and visual saliency analysis.



Haoyu Tang (Member, IEEE) received the B.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 2016 and 2021, respectively.

He is currently an Assistant Professor with the School of Software, Shandong University, Jinan, China. His research interests include machine learning and multimedia retrieval.



Yupeng Hu (Member, IEEE) received the Ph.D. degree in software engineering from Shandong University, Jinan, China, in 2018.

He is currently an Associate Professor with the School of Software, Shandong University, Jinan. His research interests include information retrieval, data mining, and explainable Artificial Intelligence (AI). Various parts of his work have been published in famous journals and forums, such as IEEE TRANSACTIONS ON IMAGE PROCESSING, *Science China Information Sciences*, and *ACM Multimedia*.

Dr. Hu has served as a PC member for ACM International Conference on Multimedia (ACM MM), Annual Meeting of the Association for Computational Linguistics (ACL), Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence (AAAI) and a reviewer for IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING and IEEE Transactions on Multimedia (TMM).



Liqiang Nie (Senior Member, IEEE) received the B.Eng. degree from Xi'an Jiaotong University, Xi'an, China, in 2009, and the Ph.D. degree from the National University of Singapore (NUS), Singapore, in 2013.

He is currently a Professor and the Dean of the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen, China. He has coauthored more than 200 articles and four books and received more than 20 000 Google Scholar citations. His research interests include multimedia computing and information retrieval.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:08:37 UTC from IEEE Xplore. Restrictions apply.